RETINA-X: A Gaze-Contingent Retinal Implant for Age-Related Macular Degeneration

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***Abstract*—Age-related macular degeneration (AMD) damages the central photoreceptors of the eye but leaves the inner retinal circuitry intact. Subretinal stimulation aims to restore vision by replacing these lost photoreceptors. However, current retinal prostheses present a clinical challenge. Camera-driven systems like PRIMA and Argus II connect perception with head movements instead of eye movements. This can make reading and visual exploration difficult. On the other hand, the gaze-contingent Alpha-AMS system achieves natural pointing through trans-scleral wiring. Unfortunately, this restricts electrode density and prevents image processing.**

RETINA-X addresses this issue by providing gaze-contingent control in a fully passive, wireless subretinal implant. The system consists of three parts: smart glasses that capture and enhance the scene, track gaze, and transmit power and data; a smart contact lens that collects power, demodulates the signal, and projects a NIR pattern with an on-lens VCSEL array; and a passive photovoltaic subretinal implant. This novel architecture aims to improve patient outcomes by providing better functional resolution through gaze contingency.

I. Introduction

A. Retinal Implants for AMD

Age-related macular degeneration (AMD) is one of the leading causes of blindness in the world – progressively destroying the photoreceptors of the central retina while leaving the inner retinal circuits, including bipolar and ganglion cells, largely intact in its early and intermediate stages (see Fig. 1) [1, 2]. Prospective therapies may take advantage of this selective degeneration: if the damaged photoreceptors can be bypassed and the surviving circuitry stimulated directly, meaningful vision may be restored (see Fig. 2) [3].

Building on this principle, subretinal photovoltaic implants convert visible or near-infrared (NIR) light into localised electrical stimulation of bipolar cells [6]. Subretinally placed implants take advantage of remaining retinal processing mechanisms by allowing stimulation of bipolar cells [3, 6]. This is a distinct advantage over epiretinal approaches, which stimulate ganglion cells, and suprachoroidal approaches, which are further away from the targeted cells [7].

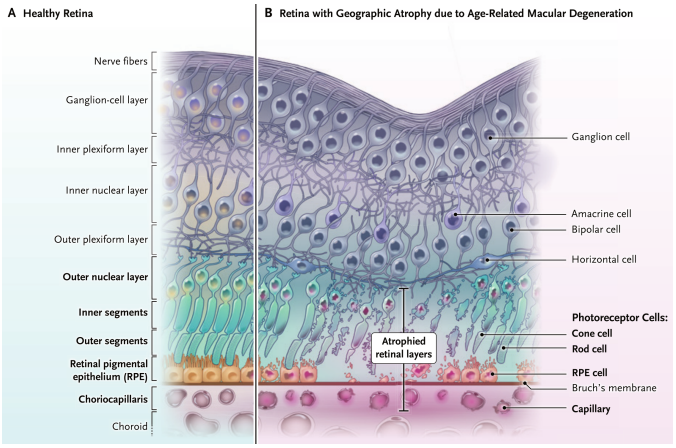


Figure 1: **(A)** Anatomy of the healthy retina and **(B)** a retina with geographic atrophy due to AMD. Taken from Holz et al. [4].

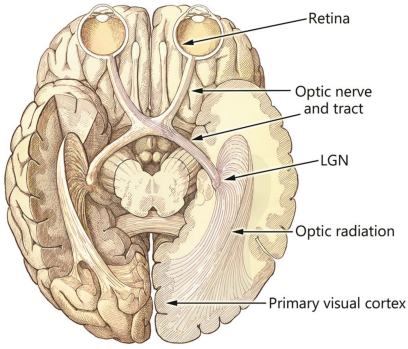


Figure 2: Implantation locations for visual prostheses. The further along the visual stream an implant is placed, the more processing it needs to compensate for. Taken from Mirochnik and Pezaris [5].

B. Clinical Relevance

Despite their promising results, most existing retinal prostheses display stimulation patterns aligned with the

head rather than eye movements [7]. In other words, they lack *gaze contingency*. As a result, patients must rotate their head or upper body to shift their visual percept. Caspi et al. [8] demonstrated that this constraint leads to slower and less precise visual exploration than natural gaze, and Paraskevoudi and Pezaris [9] confirmed that it substantially degrades reading performance compared to gaze-contingent systems.

Active photovoltaic implants such as Alpha-AMS (see Fig. 3) are able to directly produce gaze-contingent stimulation patterns from visible light without the need for an external camera [10]. However, they require trans-scleral wiring to deliver power to the photodiodes – increasing implant complexity and reducing electrode density [7]. Additionally, such implants are unable to incorporate crucial image processing algorithms for visual feature enhancement such as edges, contrast, faces, and text [11].

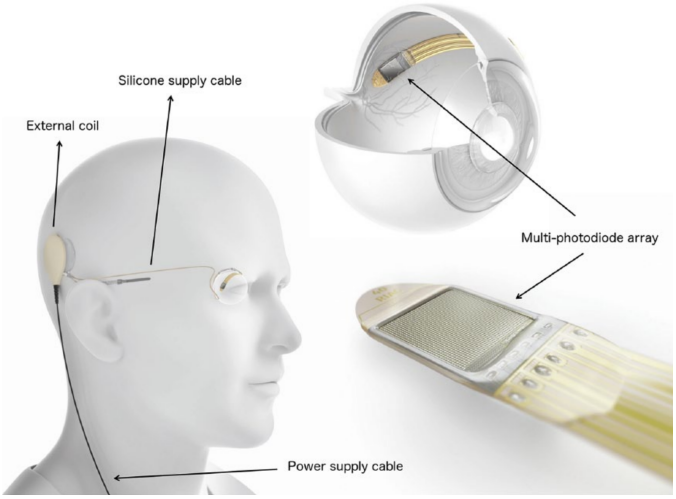


Figure 3: Illustration of Alpha-AMS. The system uses trans-scleral wiring for implant power delivery and does not rely on an external camera or processing unit. Taken from Bloch, Luo and Cruz [14].

C. Existing Treatments

A wide variety of visual prostheses for AMD have been explored clinically [12]. The further along the neural visual pathway the stimulation target is (see Fig. 2), the more the prosthesis has to compensate for the bypassed biological processing. In the case of AMD, only retinal photoreceptors are destroyed – motivating the choice of retinal implants to maximise the use of remaining pathways. Suprachoroidal implants are not discussed in detail because their greater distance from the bipolar cell layer generally increases stimulation thresholds and reduces spatial selectivity, which is less aligned with the functional resolution objective of RETINA-X [7].

Thus, the main treatment-oriented epi- and subretinal systems tested in clinical trials and with published academic results are Argus II (see Fig. 4) [11, 13], Alpha-AMS (see Fig. 3) [10], and PRIMA (see Fig. 5) [4]. Argus II functions by wirelessly transmitting power and processed data from a video camera to a processing unit sutured on the sclera. The processing unit is connected to a 6×10 epiretinal array of $200\,\mu$m-diameter micro-electrodes spaced $575\,\mu$m apart. As discussed above, Alpha-AMS functions using visible light-based photovoltaic stimulation. The implant consists of a 4×3.2 mm subretinal array of 1600 stimulation units, with each unit measuring $70\times70\,\mu$m and including a photodiode, an amplifier, and a stimulation electrode. The array is connected to an inductively-charged subdermal power supply. Finally, PRIMA functions by activating a 2×2 mm array of 378 photodiodes with NIR (880 nm) light patterns generated from a camera feed.

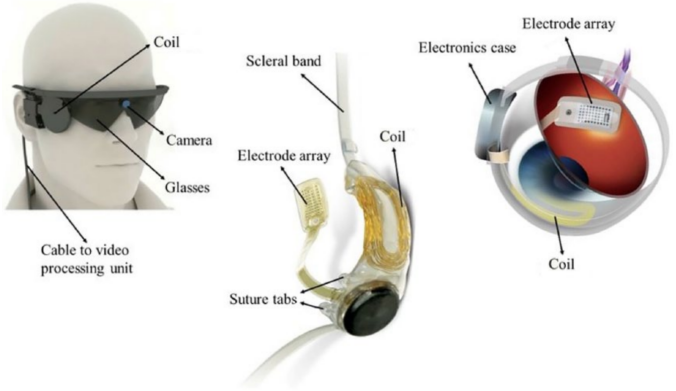


Figure 4: Illustration of Argus II. The system includes trans-scleral wiring necessary to power and drive the implant. It uses a single coil for both power and data transfer. Taken from Bloch, Luo and Cruz [14].

In addition to electronic implants, other approaches such as optogenetic therapies and direct cortical stimulation are also being studied. Current systems are compared to RETINA-X across the most relevant features in Table I.

D. Main Proposal

The main goal of the RETINA-X implant is to improve functional resolution compared to current systems. Throughout this work, functional resolution refers to the ability to generate spatially distinct percepts in the patient rather than merely increasing nominal pixel density. As discussed above, gaze contingency is a significant bottleneck for functional resolution in current systems. Higher electrode densities can be achieved with passive

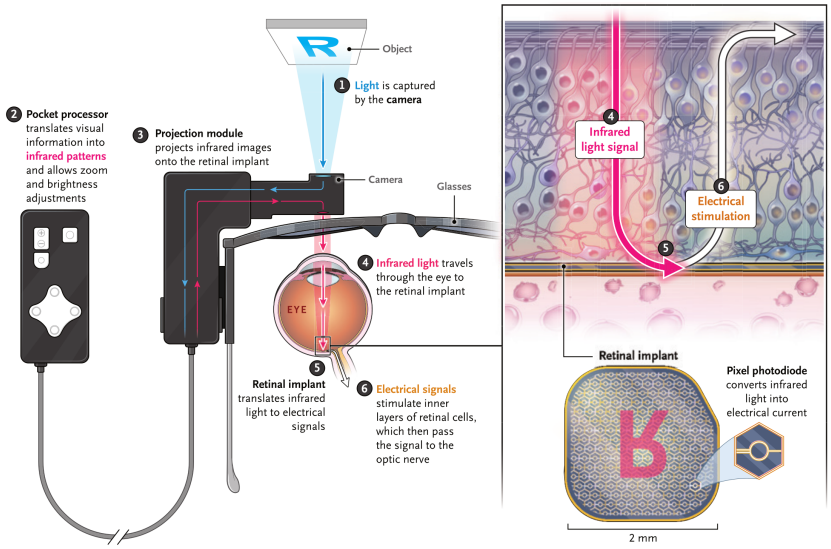


Figure 5: Illustration of PRIMA. The system processes the incoming visual stream into NIR stimulation patterns, which are projected onto a wireless subretinal photovoltaic implant. Notice the lack of any wiring for the subretinal implant. Taken from Holz et al. [4].

	Argus II	Alpha-AMS	PRIMA	**RETINA-X**
Implant site	Epiretinal	Subretinal	Subretinal	Subretinal
Stimulation	Driven electrodes	Active PV (visible)	Passive PV (NIR)	Passive PV (NIR)
Power to implant	External (cable)	External (cable)	Optical only	Optical only
Gaze-contingent	No	Yes	No	**Yes**
Image processing	Yes	No	Yes	**Yes**
Electrodes / pixels	60	1600	378	Passive PV, 3D pillar-mesh
Drive method	Camera + coil	On-chip photodiodes	External NIR goggles	On-lens VCSEL array

Table I: Comparison of clinically studied retinal prostheses with the proposed RETINA-X system.

photovoltaic arrays due to their lack of wiring [15, 4], and higher percept precision can be obtained from a subretinal implantation [7]. However, RETINA-X proposes to allow gaze-contingent control of such passive subretinal implants through a novel three-component system, depicted in Fig. 6. Together these components are designed to form a real-time adaptive system: the patient moves their eye, the system recomputes the correct gaze-contingent stimulation pattern, and projects it, all within a short enough delay. A concrete measurable endpoint for this system would be to observe a significant increase in reading performance – a highly gaze-contingent task [9] – in patients using RETINA-X.

RETINA-X is not defined by any individual component. Its novelty lies in the integration of gaze-contingent control with smart contact lens-based NIR projection and passive wireless subretinal stimulation. The comparison in Table I situates RETINA-X relative to existing systems: it is the only platform combining gaze contingency with a fully passive, wirelessly powered implant and onboard image processing. Alpha-AMS achieves gaze contingency only through a powered trans-scleral cable and cannot perform feature enhancement, while PRIMA is passive and processes images but remains head-aligned.

II. Proposed Strategy

As mentioned above, RETINA-X is built around three components which are illustrated in Fig. 6: a pair of smart glasses, a novel smart contact lens design, and a passive subretinal photovoltaic implant.

The smart glasses function as the primary energy source, computational unit, and communication hub of the RETINA-X platform. They capture the patient's surroundings with an integrated wide-angle camera and track the position of the eye via the contact lens. These gaze data allow for gaze-contingent cropping of the visual feed, which can then be processed to enhance important visual features before being wirelessly transmitted to the contact lens.

The smart contact lens hosts passive RF coils for eye tracking, a receiver coil for wireless power harvesting, an

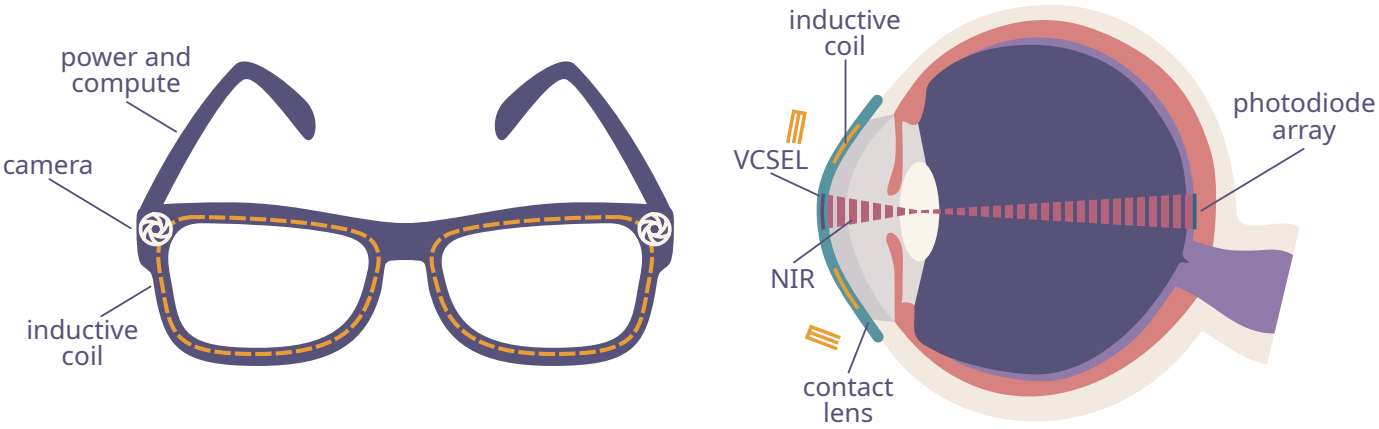


Figure 6: Overview of the RETINA-X architecture. A forward-facing camera integrated into the smart glasses captures the visual scene, which is processed by the onboard power and compute modules into a gaze-contingent stimulation pattern. Power and processed image data are wirelessly transmitted via inductive coils to the smart contact lens. The lens utilises a VCSEL to project a NIR pattern onto a passive subretinal photodiode array. This array converts the optical stimulation into localised electrical activation of surviving retinal neurons.

application-specific integrated circuit (ASIC) for signal demodulation, and a VCSEL array which projects the NIR stimulation pattern. The subretinal photovoltaic implant then functions as previously discussed, by converting the incoming NIR light into localised electrical stimulation.

Together, these components are designed to bring significant improvements to the lives of patients, most crucially: improved reading performance, better spatial navigation, enhanced object and face recognition, more natural visual exploration, and reduced cognitive load when performing visual tasks. The main expected measurable clinical endpoint of RETINA-X would be a significant increase in reading performance [9].

A. Smart Glasses

The smart glasses constitute the external control and power-delivery hub of RETINA-X – performing scene acquisition, real-time image processing, eye tracking integration, wireless power transmission, data transmission, and timing control (see Fig. 7) [16, 17]. This design choice aims to move as many components outside the contact lenses as possible, helping satisfy their multiple strict constraints including thickness, power, flexibility, heat dissipation, cost, *etc*.

A forward-facing camera integrated into the frame continuously captures the user's visual surroundings, similar to current systems like Argus II [11]. Unlike these systems however, RETINA-X uses one or multiple wide-angle cameras to record as much of the user's surroundings as possible. Thanks to the breadth of this visual stream, it can then be cropped in a gaze-contingent manner based on the data read from four RF tracking tags (see Section II-B). The cropped image stream is processed locally or on an associated compute module in real time to generate stimulation patterns. Crucially, in this step, many visual features such as contrast, faces, and text can be enhanced with transformations such as landmarking to improve visual acuity (see Fig. 8) [18, 19]. The resulting output is therefore an enhanced irradiance map to be emitted by the NIR display in the contact lens.

Wireless power delivery from the glasses to the contact lens is implemented using resonant inductive coupling [20, 21, 17]. The transmitter coil is embedded in the glasses frame, facing the eye to maximise magnetic coupling with the receiver coil [16].

The glasses also maintain a wired connection with an external control box which the user would interact with to adjust the parameters of their visual stream such as contrast, alignment, and most crucially, zoom [11, 4]. Depending on the final implementation of the glasses, this box – attached to a belt or pocket – could house the most heavy and demanding components such as the compute modules and additional batteries.

B. Eye Tracking

The eye tracking system is based on the frequency-encoded contact lens approach of Zhu et al. [22] to obtain high-precision low-latency gaze data. The key operating principle is as follows. Four passive RF resonator tags are fabricated as planar copper inductors on a thin polyimide film and encapsulated in the peripheral annular zone of the lens – leaving the central 4 mm optical zone

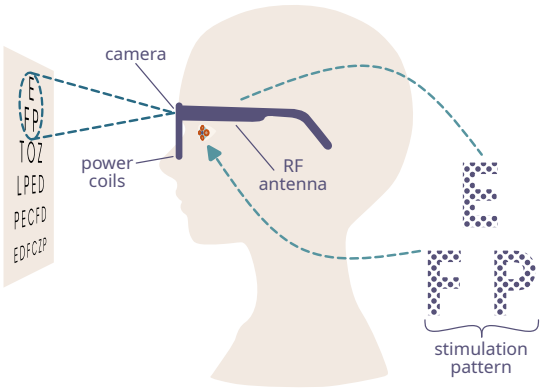


Figure 7: Proposed smart-glasses architecture for RETINA-X. The external glasses act as the acquisition, tracking, processing, and power-delivery hub of the system, combining a forward-facing camera, RF eye-tracking antenna, inductive power coils, and computational hardware to generate gaze-contingent stimulation patterns for the smart contact lens.



Figure 8: Simulated prosthetic vision of individuals expressing disgust, surprise, and happiness. The columns illustrate the following: normal vision, simulated prosthetic vision with no enhancements, enhancement via facial landmarking (0.7-implant-pixels thick in dark grey), and combined landmarking with contrast enhancement (0.3-implant-pixels thick and 50% darker than the natural feature colour for a softer effect). Taken from Goldstein et al. [18].

unobstructed. Each tag forms a passive resonant RLC circuit with a natural frequency of

$$f_{0,i} = \frac{1}{2\pi\sqrt{L_i C_i}} \qquad i \in \{1, 2, 3, 4\}$$

Distinct structural parameters enable each of the four tags to be assigned a unique resonant frequency $f_{0,i}$ spanning 500-1300 MHz [22].

The smart glasses contain a sweeping-frequency reader opposite the user's eye. When a tag is excited near resonance, it couples magnetically to the reader and modulates the reflected signal – appearing as a measurable notch in the return loss spectrum. The reader–tag coupling is characterised by mutual inductance [22]:

$$M_i = k_i\sqrt{L_i L_{\text{reader}}}$$

where $k_i \in [0, 1]$ are the coupling coefficients, which depend on the separation distance and relative orientation of the two coils. Eye movement causes these coupling coefficients to vary, which in turn varies the signal strengths received by the reader. The asymmetric layout of the radially equidistant tags can produce distinguishable differential-mode responses for both horizontal and vertical rotations. Eye closure produces a distinct common-mode amplitude drop across all four tags simultaneously [22].

Gaze coordinates are computed in real time using a sequential, time-based algorithm. Calibration of the eye tracking system is discussed in Section III-B. Once calibrated, gaze is estimated framewise by comparing consecutive temporal samples – reducing the effects of common-mode drift. Lens slippage caused by blinks and saccades shifts the resonant frequencies of the tags in a characteristic pattern, allowing the system to detect and signal the slippage direction to the user [22].

C. Power and Signal Transmission

To operate without batteries or external wiring, the contact lens must receive both stimulation pattern data and power wirelessly. RETINA-X therefore employs separate transmission channels for these functions: a 13.56 MHz inductive link for power transfer and a 2.4 GHz RF link for data transmission. The need for this separation follows from an estimate of the required image-data bandwidth.

As further discussed in Section II-D, the highest electrode count currently available in a clinically studied retinal implant is the 1600-electrode Alpha-AMS [10]. According to Goldstein et al. [18], PRIMA patients were able to distinguish approximately 7 to 8 contrast levels. Assuming these brightness values are encoded with 4 bits per electrode and transmitted at a frame rate of 30 Hz [23], the required data rate is

$$4\,\text{bit} \cdot 1600 \cdot 30\,\text{Hz} = 192\,\text{kbit}\,\text{s}^{-1}.$$

Furthermore, as discussed in Section II-D, the display would likely need to oversample the implant geometry by a factor of 4 to avoid aliasing, increasing the requirement to approximately $768\,\text{kbit}\,\text{s}^{-1}$. Optimistically assuming a data-rate-to-carrier-frequency (DRCF) ratio

of 5 % ($Q \approx 20$) [16, 17, 20], the corresponding carrier frequency would be

$$\frac{768\,\mathrm{kbit\,s^{-1}}}{0.05} \approx 15.36\,\mathrm{MHz}.$$

These assumptions already being optimistic, transmitting both power and image data through a single inductive channel would likely be challenging for RETINA-X.

A shared inductive link remains attractive because of its simplicity and space efficiency. For example, Argus II used a single internal coil operating at $3.156\,\mathrm{MHz}$ to receive both power and data (Fig. 4) [24]. This approach was sufficient for its relatively modest electrode count of 60 [11], but the substantially higher channel count targeted by RETINA-X places much greater demands on communication bandwidth. Because a single shared carrier couples data transmission to power transfer, splitting the two functions allows each link to be optimised independently.

The inductive power-transfer channel operates at $13.56\,\mathrm{MHz}$ [16]. This frequency provides an advantageous compromise among the principal constraints of ocular wireless power transfer: efficient near-field magnetic coupling, practical receiver miniaturisation, manageable passive component size, reduced sensitivity to detuning and misalignment, moderate conductor and dielectric losses, and conservative electromagnetic interaction with biological tissue [21, 20, 17].

Because the coupling between the glasses-mounted transmitter and the contact-lens receiver varies continuously during blinking, saccades, and natural eye movement, robustness is a primary design objective. The transmitter should therefore incorporate adaptive impedance matching, power regulation, and automatic frequency tracking to maintain efficient transfer under changing alignment conditions [20, 25]. Lower-frequency operation at $13.56\,\mathrm{MHz}$ is preferable to more aggressive higher-frequency regimes because it offers more predictable resonant behaviour and improved tolerance to parasitic impedance changes while remaining compatible with established biomedical wireless-power design practices.

The separate high-bandwidth data channel operates in the $2.4\,\mathrm{GHz}$ industrial, scientific and medical (ISM) band [26]. Published body-proximal and implantable transceivers at this frequency have demonstrated low-power operation together with data rates approaching the megabit-per-second range, making them substantially more suitable for image transmission than heavily modulated inductive power carriers [26].

D. Contact Lens Displays

Beyond their use in eye tracking, the smart contact lenses activate the photovoltaic electrodes of the subretinal implant. This is achieved by projecting NIR light patterns onto the implant using an integrated display. This display must fit some crucial design constraints. In the following paragraphs, these constraints are derived to be a resolution of at least $500\,\mathrm{pixel\,mm^{-2}}$, an area of at least $6\,\mathrm{mm}\times 6\,\mathrm{mm}$, a maximum intensity of at least $8.5\,\mathrm{mW\,mm^{-2}}$, and an efficiency of at least $61\,\%$. Of the candidate technologies considered, only micro-light-emitting diodes (µLEDs) [27] and vertical cavity surface emitting lasers (VCSELs) [28] appear capable of meeting the above constraints.

Both µLEDs and VCSELs generate light through electron-hole recombination. Unlike µLEDs, VCSELs incorporate a resonant optical cavity which amplifies photons through stimulated emission [29]. Both technologies have previously been integrated into smart contact lenses. For VCSELs, Robert et al. [30] integrated NIR VCSELs in contact lenses for eye tracking, while Kwon et al. [28] fabricated flexible thin-film VCSELs on biocompatible PDMS. Similarly, µLEDs have been integrated in contact-lens platforms in both academia and industry [31, 32, 33]. The suitability of these technologies, in particular VCSELs, will now be evaluated with respect to resolution, active area, irradiance, efficiency, and response time.

PRIMA [4] and Alpha-AMS [10] currently provide the highest clinically available retinal implant resolutions (Table I), with electrode densities of approximately $378\,\mathrm{pixel}/[2\,\mathrm{mm}\times 2\,\mathrm{mm}] = 94.5\,\mathrm{pixel\,mm^{-2}}$ and $1600\,\mathrm{pixel}/[4\,\mathrm{mm}\times 3.2\,\mathrm{mm}] = 125.0\,\mathrm{pixel\,mm^{-2}}$ respectively. Since perfect alignment between the contact lens and retinal implant cannot be guaranteed, the display should oversample the implant resolution to avoid aliasing. Applying the Nyquist criterion in two dimensions suggests a required display density of approximately $2^2 \cdot 125.0 = 500\,\mathrm{pixel\,mm^{-2}}$. Such densities are readily attainable for both NIR-emitting µLEDs [34, 27, 35] and VCSEL arrays [36, 28].

The macula has a diameter of $6\,\mathrm{mm}$ and clinical studies usually include AMD patients with lesion sizes in the 1.25-$18\,\mathrm{mm^2}$ range [37, 38, 39]. Using a characteristic eye focal length of $\ell \approx 17\,\mathrm{mm}$ [40] and a macular visual angle of approximately $\alpha \approx 20^\circ$ [6, 41], the corresponding projection on the corneal plane is estimated to have a radius of approximately $\ell\cdot\tan(\alpha/2) \approx 3.0\,\mathrm{mm}$. Consequently, the display should span roughly

6 mm×6 mm.

For the PRIMA implant (3 mm diameter, 75 µm pixels, 30 % duty cycle) retinal heating constraints imply a maximum NIR irradiance of approximately $8.25\,\mathrm{mW\,mm^{-2}}$. Similarly, POLYRETINA reports a maximum permissible exposure level of approximately $8.32\,\mathrm{mW\,mm^{-2}}$ at 5 Hz and 10 ms pulse duration [42]. The display should therefore be capable of producing at least around $8.5\,\mathrm{mW\,mm^{-2}}$ while allowing margin for optical losses between the cornea and retina. For NIR µLEDs, the irradiance values reported in the literature remain limited. Xue et al. [43] reported an external light emission intensity exceeding $0.4\,\mathrm{mW\,mm^{-2}}$, while substantially higher values have been demonstrated only for visible-light devices [44]. As a result, it is presently unclear whether NIR µLEDs can reliably satisfy the required irradiance target. In contrast, VCSEL arrays have demonstrated output levels largely exceeding this requirement [28].

Thermal dissipation represents a critical constraint for any implantable visual prosthesis. For a 5 mm×5 mm suprachoroidal implant, Opie et al. [45] estimated that no more than 135 mW or $5.4\,\mathrm{mW\,mm^{-2}}$ should be dissipated. Assuming this dissipated power to be the dominant source of inefficiency, combining this limit with the estimated irradiance requirement of approximately $8.5\,\mathrm{mW\,mm^{-2}}$ yields a minimum wall-plug efficiency of $\frac{8.5}{8.5+5.4} \approx 61\,\%$. This efficiency exceeds typical values reported for µLEDs [27]. By comparison, Xiao et al. [46] demonstrated a VCSEL array with a wall-plug efficiency of 74 % at room temperature, suggesting that VCSEL technology can potentially satisfy the thermal constraints of RETINA-X. Moreover, the corneal location of the display may permit more effective heat dissipation than the suprachoroidal location considered by Opie et al. [45].

Finally, pixel response time is not expected to be a limiting factor. VCSELs exhibit sub-nanosecond switching times, several orders of magnitude faster than the refresh rates required for retinal stimulation and image presentation [29]. To conclude, resolution and active-area requirements can be satisfied by either µLED or VCSEL technologies. However, the irradiance and efficiency requirements appear achievable only with VCSELs within the relevant ocular thermal limits. Combined with the maturity of NIR VCSEL technology and its demonstrated integration into contact-lens platforms, these considerations motivate the selection of a VCSEL array as the display technology for RETINA-X.

E. Subretinal Implant

The subretinal implant is the core stimulation interface of RETINA-X. As summarised in Table I, RETINA-X retains the main advantages of PRIMA-like implants: a passive, wireless, subretinal photovoltaic array activated by NIR light. However, the proposed implant introduces a key architectural improvement: a three-dimensional pillar-and-mesh electrode geometry with local return pathways.

The design of RETINA-X keeps the implanted component as simple and wireless as possible, while allowing the glasses and smart contact lens to handle image processing, gaze alignment, and NIR projection [4, 23, 15, 47]. The implant should therefore not be understood as an isolated electrode array, but as the final neural interface of a larger gaze-contingent visual restoration system. The general design logic is illustrated in Fig. 9A.

In this context, the main objective of RETINA-X is to improve functional resolution. Here, functional resolution refers to the effective ability of the implant to generate spatially distinct percepts for the patient, rather than only the nominal pixel density of the array. It depends not only on pixel pitch, but also on electrode-to-neuron distance, current confinement, and selective activation of bipolar cells without unintended stimulation of ganglion cells or axons [7]. This distinction is important because increasing the number of pixels alone does not guarantee better vision if neighbouring electrodes activate overlapping neural populations.

A central limitation of current subretinal prostheses is therefore current spread. In principle, reducing pixel size should improve spatial resolution by allowing more independent stimulation sites. Based on scaling estimates derived from PRIMA performance, reducing the pixel pitch from 100 µm to 20 µm could theoretically improve visual acuity from approximately 20/420 to around 20/80 [48]. However, this theoretical gain can only be achieved if each pixel activates a sufficiently distinct group of retinal neurons. In AMD patients, subretinal debris and imperfect retinal apposition can increase the distance between the implant and the bipolar cell layer. This weakens stimulation, raises activation thresholds, and promotes lateral spread of the electric field before it reaches the target cells [19, 48]. As a result, adjacent pixels may no longer act independently, and the perceptual benefit of miniaturisation is partly lost, as illustrated by the stimulation patterns in Fig. 9B.

RETINA-X addresses this bottleneck by proposing a three-dimensional photovoltaic interface rather than

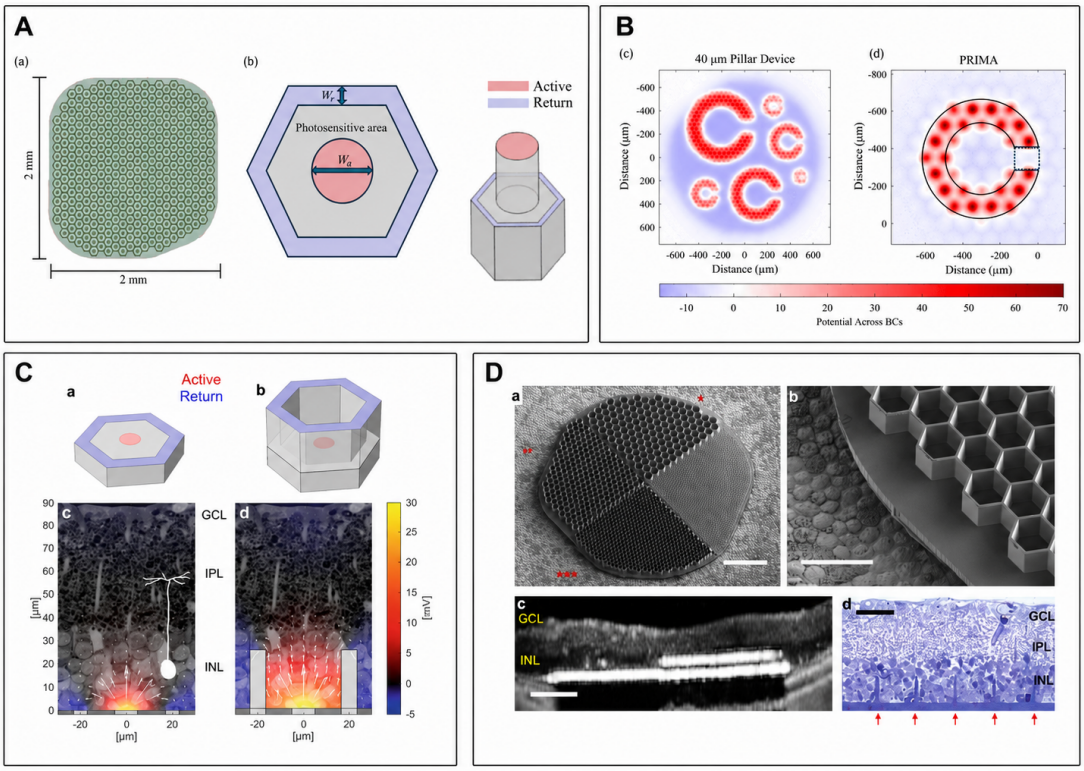


Figure 9: Retained RETINA-X subretinal implant concept and possible future extension. **(A)** PRIMA-like passive photovoltaic architecture with a hexagonal pixel array and local active/return electrode layout. **(B)** Simulated stimulation patterns comparing a pillar-based device with the current PRIMA design, illustrating the relevance of spatial current confinement. **(C)** Planar versus three-dimensional pillar-and-mesh electrode geometries with local return pathways; this corresponds to the retained RETINA-X implant concept. **(D)** Honeycomb-shaped subretinal interface shown as a possible future extension, not as part of the baseline RETINA-X design. Panels **(A)** and **(B)** adapted from Jensen et al. [48]; panels **(C)** and **(D)** adapted from Flores et al. [49].

a denser planar array. In the selected pillar-and-mesh architecture, active electrodes are elevated on small pillars, bringing the stimulation source closer to the target bipolar cells. Surrounding return electrodes form a local mesh around each photovoltaic pixel, helping to confine the electric field and reduce lateral current spread. This geometry is expected to preserve stimulation strength and spatial contrast even when pixel pitch is reduced, while maintaining preferential activation of bipolar cells [48]. Compared with a planar geometry, the three-dimensional structure shown in Fig. 9C therefore directly addresses the physical bottleneck that prevents smaller pixels from automatically becoming functionally useful pixels.

Mechanical integration is also essential for functional resolution. The retina is soft, curved, and fragile, whereas microelectronic implants are often relatively rigid. Poor conformity between the implant and the retinal surface increases electrode-to-neuron distance and can further degrade stimulation selectivity. For this reason, the RETINA-X implant should ideally use a mechanically hybrid design: rigid enough to allow controlled surgical insertion and stable positioning beneath the macula, but sufficiently conformable at the retinal interface to improve apposition to the retinal curvature [48, 50]. This balance is important because excessive flexibility could make the device difficult to insert, position, and stabilise, whereas excessive rigidity could compromise tissue contact and increase stimulation thresholds [50, 51].

The subretinal implant is the most invasive component of RETINA-X and would therefore rely on established subretinal surgical procedures. Implantation would be performed by a specialised vitreoretinal surgeon, beginning with pars plana vitrectomy followed by the creation of a localised subretinal bleb to allow insertion of the device beneath the macula [6, 52]. After placement, close and stable apposition between the implant and the retinal tissue would be essential, since electrode-tissue

distance directly influences activation thresholds, current spread, and stimulation precision [6, 42]. If a partially conformable substrate is used, the surgical principle would remain similar to current subretinal implantation, but the handling of the device would need to be carefully optimised to balance improved tissue contact with surgical stability [50, 51].

Because retinal morphology varies between patients, procedures such as optical coherence tomography could be performed post-operatively to estimate local retinal thickness, implant position, and electrode-to-neuron distance [53, 54]. These parameters could then inform initial RETINA-X calibration procedures.

Post-operative adaptation would also be an important part of the RETINA-X strategy. Because the implant operates within a gaze-contingent system, the projected NIR pattern must remain aligned with the implanted array during eye movements. Initial alignment would rely on the RF-resonator eye-tracking system described in Section II-B, while stimulation parameters could later be refined through psychophysical testing, including percept localisation, contrast discrimination, and reading-like tasks [18, 48]. Over time, patients would also need to learn how to interpret the artificial stimulation patterns generated by the device. This neuroadaptation process should be considered part of the functional performance of the system rather than a secondary rehabilitation step, since patients are expected to gradually extract more meaningful visual information from initially unfamiliar prosthetic percepts [23, 6].

Beyond the design choices retained for the initial RETINA-X implant, two possible future extensions may be considered. First, honeycomb-shaped electro-neural interfaces are included in Fig. 9D as a possible next-generation architecture. Such interfaces are conceptually attractive because their three-dimensional microstructure could bring retinal neurons into closer contact with the stimulating surface and may improve robustness to anatomical variability [49, 55]. However, they would also introduce additional fabrication, surgical, and translational uncertainty, particularly regarding tissue integration, insertion beneath the macula, long-term biocompatibility, and reproducible neuron-interface apposition. Second, optogenetic sensitisation of bipolar cells could theoretically reduce stimulation thresholds and improve implant efficiency [56]. However, this would require an additional gene-therapy component, raising substantial safety, regulatory, and patient-selection challenges [57, 58]. For these reasons, neither honeycomb interfaces nor optogenetics are retained in the initial RETINA-X design. Instead, the baseline system remains a PRIMA-inspired passive photovoltaic platform with local pillar-and-mesh current confinement, while these approaches are best considered future extensions if the initial passive platform is first validated.

III. Implementation and Methodology

A. Theoretical Timeline

A conceptual timeline of the major milestones for the development of RETINA-X is depicted in Fig. 10. The translation of RETINA-X into clinical applications would likely be very challenging due to its potential risks, its need for multiple technological developments, and its three interdependent components. The subretinal implant is the most invasive component of RETINA-X, thus, the first translational step should not necessarily require implanting a new device. A more realistic strategy would be to test the RETINA-X glasses and contact lens in patients who already carry a PRIMA or PRIMA-like passive photovoltaic implant, provided that optical, regulatory, and ethical compatibility is established. This would allow the system-level hypothesis of RETINA-X to be evaluated while avoiding an additional intraocular surgical procedure. In this configuration, the existing implant would serve as a clinically validated stimulation target, while the new components would test whether gaze-contingent NIR projection, eye tracking, and image preprocessing can improve the functional use of prosthetic vision. Such a study would not validate the future improved implant geometry itself, but it would answer a crucial first question: whether the three-device architecture can operate effectively in clinical settings. Only after this compatibility step would it be justified to move toward a dedicated next-generation RETINA-X implant.

RETINA-X would likely need to satisfy many standards in the pre-clinical phase, the most important of which are shown in Fig. 10. These standards centre around the packaging of the contact lens, the risks involved in using high-intensity NIR lasers (VCSELs), thermal and biocompatibility concerns, and the security of the software and wireless communication. Extensive bench testing would thus be required, such as thermal finite element modelling (FEM) [59], testing in cadavers and enucleated eyes [45, 60], and implant fidelity simulations [48, 19].

An *in vivo* testing phase would most likely be necessary to assess safety. Rabbits [61, 32, 62] and pigs [60] are the most commonly employed animal models

to evaluate smart contact lens safety. Rabbits have very sensitive corneas which are ideal for such experiments, but pig eyes are much closer in size and anatomy to human eyes. Such animal models would provide a platform to test contact lens thermals, fit, thickness and oxygen transfer, as well as biocompatibility.

Despite the glasses and contact lens being modular and compatible with existing implants, notably PRIMA, it may still be necessary to evaluate efficacy before human testing. Non-human primates (NHPs) could thus be employed for this. All components of RETINA-X would have to be applied to the NHPs, including a subretinal implant. Tests such as the one presented by Killian et al. [63] could be adapted and/or simplified to measure the efficacy of the system. Simplification may be needed since, for the case of Killian et al. [63], the NHPs had to be trained for months to complete the task with simulated phosphene vision, and this would likely be even more challenging for RETINA-X.

Finally, for the approval phase, RETINA-X is clearly a Class III system [64, 65, 66]. This implies that FDA and EMA approvals would be strict. The SENSIMED Triggerfish [67] can serve as an example to ground the proposed clinical translation pathway of the RETINA-X smart contact lens. The Triggerfish system is one of the only smart contact lens systems which is currently an approved and available therapy. A summary of its translation milestones is depicted in Fig. 11.

The Triggerfish is a disposable contact lens designed to be worn for a 24 h period. It contains a microprocessor, an antenna, and two strain gauges which detect changes in corneal shape to measure intraocular pressure (IOP). An adhesive coil attached on the skin around a patient's eye powers the contact lens and receives the strain gauge data to be recorded.

This system is clearly simpler than RETINA-X, not only in terms of the number of components but also in terms of the complexity and associated risk of the individual components (*e.g.* no high-intensity NIR VCSEL array). The RETINA-X timeline would thus likely be longer and more demanding. However, it can still serve as a useful case study for RETINA-X.

B. Core Clinical Challenges

Central engineering challenges have already been discussed above. However, RETINA-X is faced with several clinically-focused challenges, such as thermal safety. The established safety standard for active medical implants (ISO 14708-1 [68]) recommends a temperature rise limit

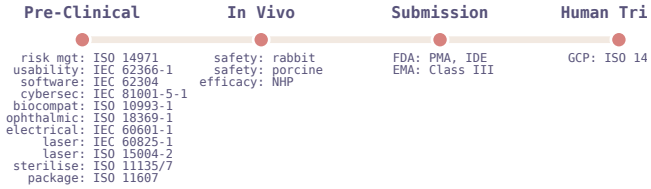


Figure 10: Conceptual timeline of the major milestones for the development of RETINA-X. Data were acquired from ISO [68], IEC [69], FDA [64, 65] and EMA [66].

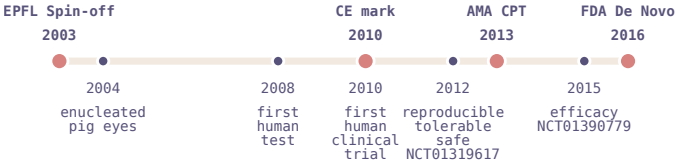


Figure 11: Timeline of the clinical translation process of the SENSIMED Triggerfish. Data were acquired from SENSIMED [70], Leonardi et al. [60], Pitchon et al. [71], Faschinger and Mossböck [72], Mansouri et al. [73], Mansouri, Weinreb and Liu [74] and Dunbar, Shen and Aref [67].

of 2 °C. Additionally, the guidelines of the International Commission on Non-Ionizing Radiation Protection (ICNIRP) state the specific absorption rate (SAR) should not exceed $2\,\mathrm{W\,kg^{-1}}$ [16]. Component-specific estimations as in Section II-D can be performed, however methods such as finite element modelling (FEM) [59] and thermal testing in cadavers and enucleated eyes [45, 60] would be necessary to assess the thermal safety of RETINA-X.

Another clinical challenge, related to eye tracking (see Section II-B), is calibration. Once the contact lens has been applied and the glasses are being worn, the gaze-tracking system must be calibrated to map the RF resonator readings to the true gaze angle. The swirling calibration routine detailed by Zhu et al. [22] is not compatible with blind patients since the subject has to fixate on visual targets – requiring functional vision [75]. Several approaches exist which allow calibration independent of subject perception such as with stereoscopic cameras combined with computer vision [75]. As reviewed by Paraskevoudi and Pezaris [9], it is expected that such approaches could then be fine-tuned by combining them with perceptual processes such as pointing tasks, staircase saccades, vestibulo-ocular reflex (VOR), smooth pursuit, *etc*.

The calibrated system would also need to minimise the frequency of recalibration during use. As the glasses are not rigidly fixed to the user's head, slippage or shaking may occur, in addition to the contact lens slippage mentioned in Section II-B. For slippage of the glasses, the user may simply need to be notified, *e.g.* with a

beep, to readjust their glasses. Shaking however may be able to be compensated for through integrated inertial measurement units (IMUs) in the glasses.

Similar to heat, another challenge which accumulates between components is latency. The most temporally demanding components of RETINA-X are likely the initial visual processing paired with the signal transmission and demodulation. A latency of 60 ms has been found to be the maximum limit before humans report lag in gaze contingency [76]. Additionally, current clinical implants such as PRIMA apply a stimulation frequency of 30 Hz to provide steady perception. A reasonable target latency for RETINA-X could thus be 30 ms, such that each frame is processed independently while still meeting the above constraints.

Another core challenge the development of RETINA-X would have to overcome is contact lens thickness. Indeed, contact lens thickness not only impacts flexibility and comfort, but also oxygen diffusion [62]. The cornea being avascular, it requires access to oxygen from the air. For instance, the SENSIMED Triggerfish discussed in Section III, which can be worn for at least 24 h, is composed of silicone and has a central thickness of 585 µm [67]. The thickness of the RETINA-X smart contact lens would crucially determine how frequently the lens would have to be removed.

This directly links to a host of challenges centred around usability. In particular, applying and calibrating the contact lenses may be difficult or unfeasible for a blind user. Thus, they would likely require a caretaker to accomplish these tasks. Depending on the frequency of reapplication and the durability of the lens, these may also need to be regularly cleaned to avoid infection. Additionally, the cost of a RETINA-X contact lens would have to be weighed against its durability to determine economic viability of the RETINA-X treatment for patients.

IV. Discussion

This work proposes RETINA-X as an extension of existing retinal prosthesis strategies. Since AMD primarily damages the photoreceptors while leaving much of the inner retinal circuitry intact (Fig. 1) [1, 2], a subretinal implant remains a logical approach, and RETINA-X builds on it by combining passive photovoltaic stimulation with a gaze-contingent optical interface.

The central strength of the system is the integration of its three components (Fig. 6), each addressing a specific limitation of existing prostheses: the implant provides stimulation close to the remaining retinal circuitry (Section II-E), the contact lens projects a NIR pattern onto it (Section II-D), and the glasses handle image preprocessing and wireless power and data transfer (Section II-A). A summarised illustration of these processes is provided in Fig. 12. A key finding of this design evaluation is that gaze contingency is not merely an additional feature but a central design motivation: by allowing the stimulus to follow gaze rather than head position [7], RETINA-X could support faster scene exploration, improved reading, better navigation, and more natural recognition of objects and faces.

These advantages map onto the everyday tasks that most affect autonomy in patients with central vision loss. Gaze-contingent reading is the clearest example, as full gaze contingency improves reading performance relative to head-steering alone [9], while image-enhancement modules could further support recognition of faces and text. Improved alignment between percept and eye movement may also benefit navigation and object localisation, for which head-aligned systems require slower compensatory head motion [8]. These outcomes: reading speed, face recognition, navigation, and daily independence, define the clinically meaningful endpoints against which the system should ultimately be evaluated.

The constraints posed by the individual components of RETINA-X, in particular those of the smart contact lenses, were assessed for feasibility. Current technologies, such as VCSELs for the NIR displays, show promise and suggest feasibility for the RETINA-X system.

However, the feasibility of individual components does not guarantee feasibility of the complete system. RETINA-X depends on precise coordination between the implant, contact lens, eye tracking, and glasses, and challenges such as heat, latency, gaze calibration, lens slippage, contact lens thickness, wireless power delivery, and long-term daily use remain central limitations, since instability in any one part could degrade the entire system. The translational pathway must therefore validate the system step by step, thermal modelling for heat-related safety, visual acuity tasks such as the Landolt C test for functional improvement, rabbit studies for ocular safety, and non-human primate experiments for gaze-contingent behaviour (Section III-B), because these technologies must be validated both separately and together.

Introducing such a device into daily life also raises ethical considerations. Because RETINA-X combines an implant, contact lens, smart glasses, eye tracking, and image preprocessing, issues such as informed consent, long-term safety, dependence on external hardware, data

privacy, and accessibility are important. The goal is not only to improve visual performance, but to ensure the technology remains safe, transparent, and useful for those who rely on it.

Overall, the analyses in this work support RETINA-X as a promising but complex retinal prosthesis concept. Its main advantage, the integration of high-resolution subretinal stimulation, an efficient NIR display, gaze tracking, and visual preprocessing, is also its main challenge: RETINA-X can be clinically meaningful only if these components function as one stable, comfortable, and reliable visual interface.

V. Conclusion

RETINA-X proposes a gaze-contingent retinal prosthesis system designed to improve the functional resolution and usability of artificial vision in patients with AMD. By combining a subretinal photovoltaic implant, a NIR VCSEL contact lens display, eye tracking and smart glasses for visual preprocessing, the system aims to preserve the advantages of subretinal stimulation while making prosthetic vision more natural and responsive to eye movement.

The results suggest that this approach is promising because it addresses several limitations of existing retinal implants, especially the lack of gaze-contingent visual input and the limited functional use of low-resolution percepts. The feasibility analyses propose strong candidate technologies which could meet the technical and clinical requirements of RETINA-X.

However, RETINA-X remains a complex system with important engineering, clinical, and ethical challenges. Thermal safety, latency, calibration, lens comfort, long-term reliability, patient accessibility, and informed consent must all be carefully evaluated before clinical translation. Overall, RETINA-X represents a plausible next step in retinal prosthesis development, but its success will depend on proving that its integrated components can work safely, comfortably, and meaningfully in daily life.

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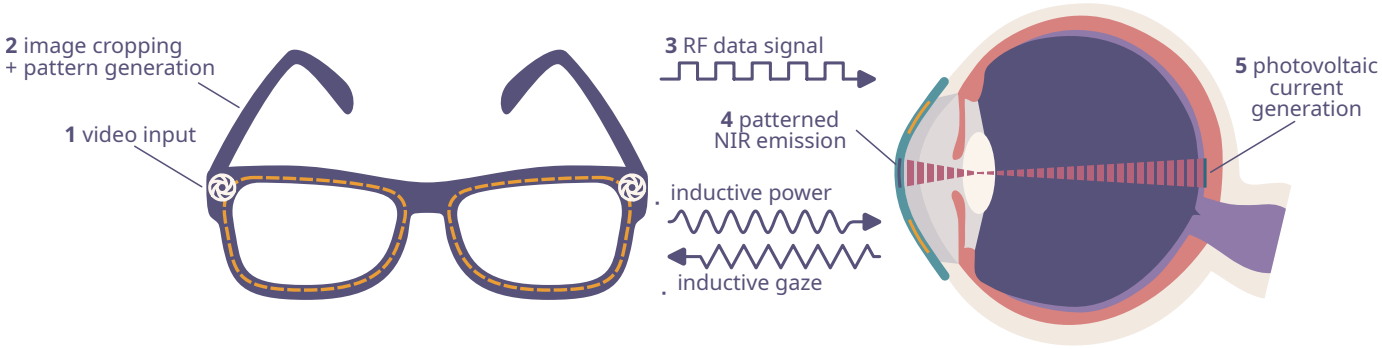


Figure 12: Summary illustration of RETINA-X. Each major step is numbered and continuous processes are denoted by cyclic arrows.

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